Emily K. Johnson

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PROFILE

Health economist and population health researcher, trained as a statistician and health data scientist. My work measures population health and its economic consequences using administrative data at national scale, with methods spanning register-based causal inference, machine learning, and Bayesian modelling. Before my PhD I spent four years as a Research Scientist at the Institute for Health Metrics and Evaluation and two years at Epic Systems. I collaborate with research groups at the University of Oxford, the Institute for Health Metrics and Evaluation, Johns Hopkins University, and Yale University.

APPOINTMENTS

University of Southern Denmark — Odense, DK
Assistant Professor, Danish Centre for Health Economics — August 2026 – Present
Visiting Fellow, Danish Institute for Advanced Study — 2026 – Present

EDUCATION

University of Southern Denmark — Odense, DK
Doctor of Philosophy in Health Economics — 2026
- Affiliations: Danish Centre for Health Economics, Danish Institute for Advanced Study
- Supervised by Angela Chang, DSc, Liza Sopina, PhD, and Harsh Parikh, PhD

Johns Hopkins University — Baltimore, MD, USA
Visiting Graduate Student — May 2025 – March 2026

London School of Economics — London, UK
Master of Science in Health Data Science (Distinction) — July 2023
- **Prize for Excellence in MSc Health Data Science**, highest average 2022/23
- Program Representative, Student Staff Liaison Committee

University of Edinburgh — Edinburgh, UK
Postgraduate Certificate in Epidemiology — September 2023

University of Washington — Seattle, WA, USA
Non-matriculated postgraduate student during full-time employment — 2019 – 2022

Rice University — Houston, TX, USA
Bachelor of Arts in Statistics, Minor in Global Health Technologies (GPA 3.52) — May 2017

PUBLICATIONS

Peer-reviewed articles

Author lists longer than 8 names are abbreviated.

1. Parikh H, McCormick T, **Johnson EK**, Hickey L, Ranney ML, Mukherjee B. The Epidemiology of Artificial Intelligence. *Nature Health*. 2026.
2. **Johnson EK**, Parikh H, Ettman CK, Ge G, Sopina L, Chang AY. Lasting Income Costs of Mental and Physical Illness. *JAMA Health Forum*. 2026.
3. Chang AY, **Johnson EK**, Bolongaita S, et al. From Sex Differences to Sex Inequalities in Life Expectancy: A Cross-Country Observational Benchmarking Analysis. *PLOS Medicine*. 2025.
4. **Johnson EK**, Parikh H, Olsen KR, Chang AY, Sopina L. Breast Cancer and Income Loss in Denmark: Heterogeneous Outcomes and Longitudinal Effects. *Nature Communications*. 2025.

5. Lescinsky H, Sahu M, Beauchamp M, ..., **Johnson E**, ..., Milstein A. Exploring State-Level Change in Health Care Value Over Three Decades in the United States, 1991–2020. *Health Services Research*. 2025.
6. Aagaard A, Röttger R, **Johnson EK**, Olsen KR. Comparing the Predictive Performance of Diabetes Complications Using Administrative Health Data and Clinical Data. *Scientific Reports*. 2025.
7. Dieleman JL, Beauchamp M, Crosby SW, ..., **Johnson EK**, ..., Murray CJL. Tracking US Health Care Spending by Health Condition and County. *JAMA*. 2025.
8. Mokdad AH, Bisignano C, Hsu JM, ..., **Johnson EK**, ..., Murray CJL. The Burden of Diseases, Injuries, and Risk Factors by State in the USA, 1990–2021: A Systematic Analysis for the Global Burden of Disease Study 2021. *The Lancet*. 2024.
9. Norheim OF, Chang AY, Bolongaita S, ..., **Johnson EK**, ..., Peto R. Halving Premature Death and Improving Quality of Life at All Ages: Cross-Country Analyses of Past Trends and Future Directions. *The Lancet*. 2024.
10. Naghavi M, Ong KL, Aali A, et al. Global Burden of 288 Causes of Death and Life Expectancy Decomposition in 204 Countries and Territories and 811 Subnational Locations, 1990–2021: A Systematic Analysis for the Global Burden of Disease Study 2021. *The Lancet*. 2024.
11. Paget J, Staadegaard L, Wang X, ..., **Johnson EK**, ..., Spreeuwenberg P. Global and National Influenza-Associated Hospitalisation Rates: Estimates for 40 Countries and Administrative Regions. *Journal of Global Health*. 2023.
12. Roberts NLS, **Johnson EK**, Zeng SM, et al. Global Mortality of Snakebite Envenoming between 1990 and 2019. *Nature Communications*. 2022.
13. **Johnson EK**, Wojtesta MA, Crosby SW, et al. Varied Health Spending Growth Across US States Was Associated With Incomes, Price Levels, And Medicaid Expansion, 2000–19. *Health Affairs*. 2022.
14. **Johnson EK**, Sylte D, Chaves SS, et al. Hospital Utilization Rates for Influenza and RSV: A Novel Approach and Critical Assessment. *Population Health Metrics*. 2021.
15. Li Y, **Johnson EK**, Shi T, et al. National Burden Estimates of Hospitalisations for Acute Lower Respiratory Infections Due to Respiratory Syncytial Virus in Young Children in 2019 among 58 Countries: A Modelling Study. *The Lancet Respiratory Medicine*. 2021.

Works in progress

- **Johnson EK**, Parikh H, Ettman C, Ge G, Sopina L, Chang AY. Hospital Diagnosed Disease and Family Formation: Evidence from Danish Register Data. In preparation.
- **Johnson EK**, Lindberg A, Marin AO, Riumano Herl C, Baravelli CM, Chang AY, Emmanuel E. Who dies early? Predicting and Describing Premature Mortality using Danish Register Data. In preparation.

Commentary

- Chang AY, **Johnson EK**. Confronting the Systematic Invisibility of Women's Health. *The Lancet Obstetrics, Gynaecology, & Women's Health* (commentary). 2025.
- **Johnson EK**, Dieleman JL. Standardizing State Health Spending. *Health Affairs Forefront* (blog post). 2022.

RESEARCH EXPERIENCE

University of Southern Denmark, Danish Centre for Health Economics Odense, DK

Experienced Burden of Disease Project (PhD research) 2023 – 2026

- Measuring the economic, social, and health loss attributable to breast cancer, stroke, alcohol use disorder and depression over an 18-year period in Denmark using administrative health, education, and economic data.

Predicting Early Mortality Project (PhD research) 2023 – 2026

- Identifying and describing individuals who die early in Denmark using machine learning and decomposition methods and over 2000 individual-level health, economic, demographic and social characteristics.

Contributor, International Network on Comparative Resource Use 2024 – Present

- Comparing hospital resource use by disease, age, sex, and comorbidity status in Denmark to 16 other high-income countries globally in collaboration with the University of Oxford and Institute for Health Metrics and Evaluation.

University of Washington, Institute for Health Metrics and Evaluation Seattle, WA
Research Scientist I, US Disease Expenditure Project 2021 – 2023
- Estimated US county and state-level disease-specific health expenditure by payer and type of care from 2000-2019, using billions of rows of administrative data and econometric and small area estimation methods.

Research Scientist I, Global Burden of Disease (GBD), Clinical Informatics 2019 – 2021
- Estimated incidence and prevalence of disease through Bayesian mixed effects modeling for 85 diseases and injuries using hospital administrative data from over 30 countries. Modeled outputs are a key input to Global Burden of Disease morbidity modeling.

PRESENTATIONS

Invited seminars

Using AI for quantitative research

- Yale University, Upcoming
- Aarhus University

Conference presentations

Johnson EK, Parikh H, Ettman C, Ge G, Sopina L, Chang AY. *Lasting Income Costs of Illness: a Danish Register Study of Mental and Physical Disorders.*

- Årsmøde i Dansk Selskabe i Sundhedsøkonomi, Odense DK. September 2025. Oral presentation.
- European Health Economics PhD Conference, Brussels BE. September 2025. Oral presentation.
- Nordic Health Economics Study Group, Oslo NO. August 2025. Oral presentation.
- International Health Economics Association Congress, Bali ID. July 2025. Oral presentation.

Johnson EK, Parikh H, Sopina L, Olsen KR, Chang AY. *Economic Burden of Breast Cancer in Denmark: Longitudinal Estimates and Heterogeneous Effects.*

- International Health Economics Association Congress, Bali ID. July 2025. Oral presentation.
- Nordic Epidemiology Conference, Copenhagen DK. May 2025. Oral presentation.

Johnson EK, Wojtesta M, Crosby SW, Duber HC, Jun E, Lescinsky H, Nguyen P, Sahu M, Thomson A, Tsakalos G, Weil MS, Haakenstad A, Mokdad A, Murray CJL, Dieleman JL. *Variation in US State Health Spending Grows and Is Associated with Income, Health System Capacity, Insurance Coverage and Market Concentration.*

- American Society of Health Economists Conference, Austin US. June 2022. Oral presentation.

TEACHING

University of Southern Denmark Esbjerg, DK
Lecturer, examiner, and course administrator, Global Health Policy graduate course Spring 2024
MSc thesis supervision, MSc Data Science, 4 students Spring 2024, 2025

Rice University Houston, TX
Teaching assistant, Global Health Technology undergraduate course Spring 2017
Teaching assistant, Probability and Statistics undergraduate course Fall 2016

OTHER PROFESSIONAL EXPERIENCE

Epic Systems Corporation Madison, WI
Technical Services Manager 2018 – 2019
- Supervised technical services analysts on the Healthy Planet value-based care and risk management application. Conducted performance evaluations, assigned customer projects, and coached new hires in technical skills.

Technical Services Analyst 2017 – 2018
- Managed project to integrate claims data sent from payer contracts into the Epic data warehouse and patient medical records for clinical decision-making. Improved quality of care metrics by 35% with additional data.

- Supported an implementation of Epic software for a value-based care management and pediatric diabetes initiative in California. Built and tested tools for reporting, patient tracking, and care planning used by 53 care managers and health coaches, who enrolled 5000 new patients.

SKILLS

- 8 years experience in R (data.table, dplyr, arrow, ggplot, plotly), 6 in Python (pandas, seaborn, matplotlib, arrow), and 3 in SQL.
- Experienced with Github, repository management, and version control.
- Experienced with parquet, Spark, Arrow, distributed and cluster-based computation, and management of big data on the scale of billions of rows and tens of terabytes of memory.
- Familiar with Claude Code, Elicit, and agentic AI workflows.
- Conversant in French, Spanish and Danish.